

# Air Quality Prediction

MULTI-SOURCE REMOTE SENSING & HYBRID DEEP LEARNING AND  
IMPLEMENTATION OF THE MAST-NET FRAMEWORK



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# Problem Statement

- **Health Crisis:** Pollution linked to millions of respiratory-related deaths annually.
- **Infrastructural Gap:** Ground stations are sparse, leaving 70% of urban areas unmonitored.
- **Data Staticity:** Current models view pollution as "snapshots," ignoring temporal flow.
- **Policy Need:** Governments lack high-resolution, predictive tools for real-time alerts.
- **Standardization:** Urgent need for IEEE-standard uncertainty quantification in AI results
- **Real-World Urgency:** Recent city-wide "Red Alerts" highlight that static monitoring fails to provide the 48-hour lead time necessary for emergency public health actions.

# Literature Review

- **Foundation AQ-Net (2023)** : Proved multimodal AI can estimate NO<sub>2</sub> via Sentinel-5P.
- **Evolution (2026)** : Shifted from "Simple CNNs" (Vision) to "Spatio-Temporal Hybrids" (Vision + Memory).
- **Current**: The reference paper introduces **MAST-Net**, focusing on Attention mechanisms and Triple-Satellite (**MODIS/Landsat/Sentinel**) fusion.

# Methodology

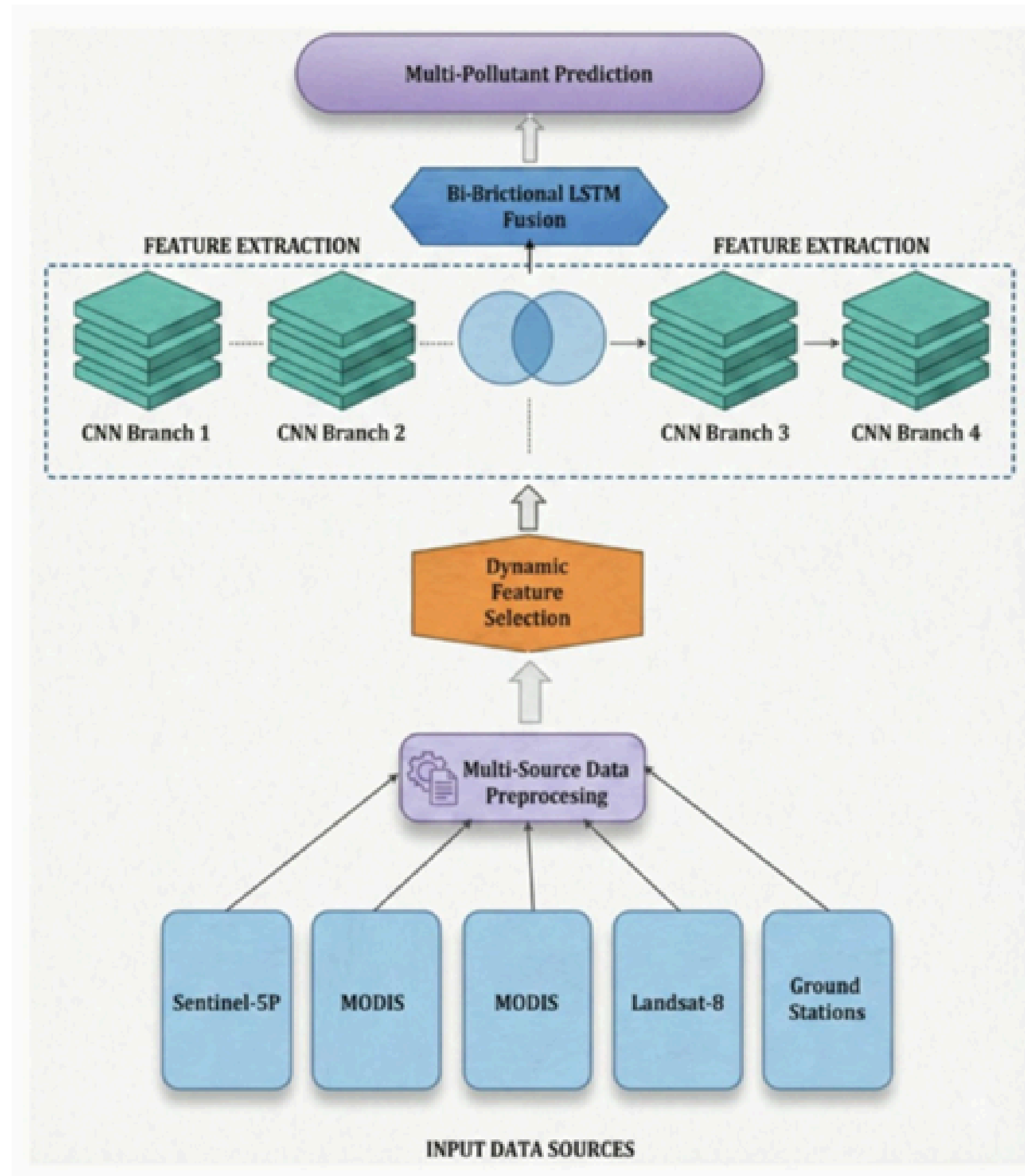
## The MAST-Net Framework

- **Hybrid Core:** Merges CNNs (Spatial/Maps) with Bi-LSTM (Temporal/Memory) to understand both "where" and "when" pollution moves.
- **Multi-Modal Branching:** Uses 4 parallel streams to process Sentinel, MODIS, and Landsat data separately, preventing data "blurring" and preserving detail.
- **DFS Gating:** A smart Sigmoid filter that automatically ignores "garbage" data (like clouds or sensor errors) in real-time.
- **8-Head Attention:** The "Brain" that synchronizes maps with weather timelines, focusing only on the most critical environmental drivers.
- **Uncertainty-Aware (UQ):** Outputs a Probabilistic Score (Mean + Variance). It doesn't just guess; it tells you exactly how confident it is in the prediction.

# Data Integration

**Multi-Satellite Fusion** : Moves beyond single-source sensing by integrating three distinct satellite platforms:

- **Sentinel-5P (TROPOMI)**: High-sensitivity mapping of trace gases (NO<sub>2</sub>, O<sub>3</sub>, SO<sub>2</sub>).
  - **MODIS (Terra/Aqua)**: Provides Aerosol Optical Depth (AOD) to track dust, smoke, and particulate matter (PM<sub>2.5</sub>).
  - **Landsat-8**: Delivers high-resolution surface data to monitor urban density and thermal patterns.
1. Branch 1 (Sentinel): Extracts chemical signatures and gas concentrations.
  2. Branch 2 (MODIS): Tracks aerosol density and particulate flow.
  3. Branch 3 (Landsat): Analyzes thermal patterns and urban morphology.
  4. Branch 4 (Auxiliary): Integrates ground-truth calibration and meteorological trends.



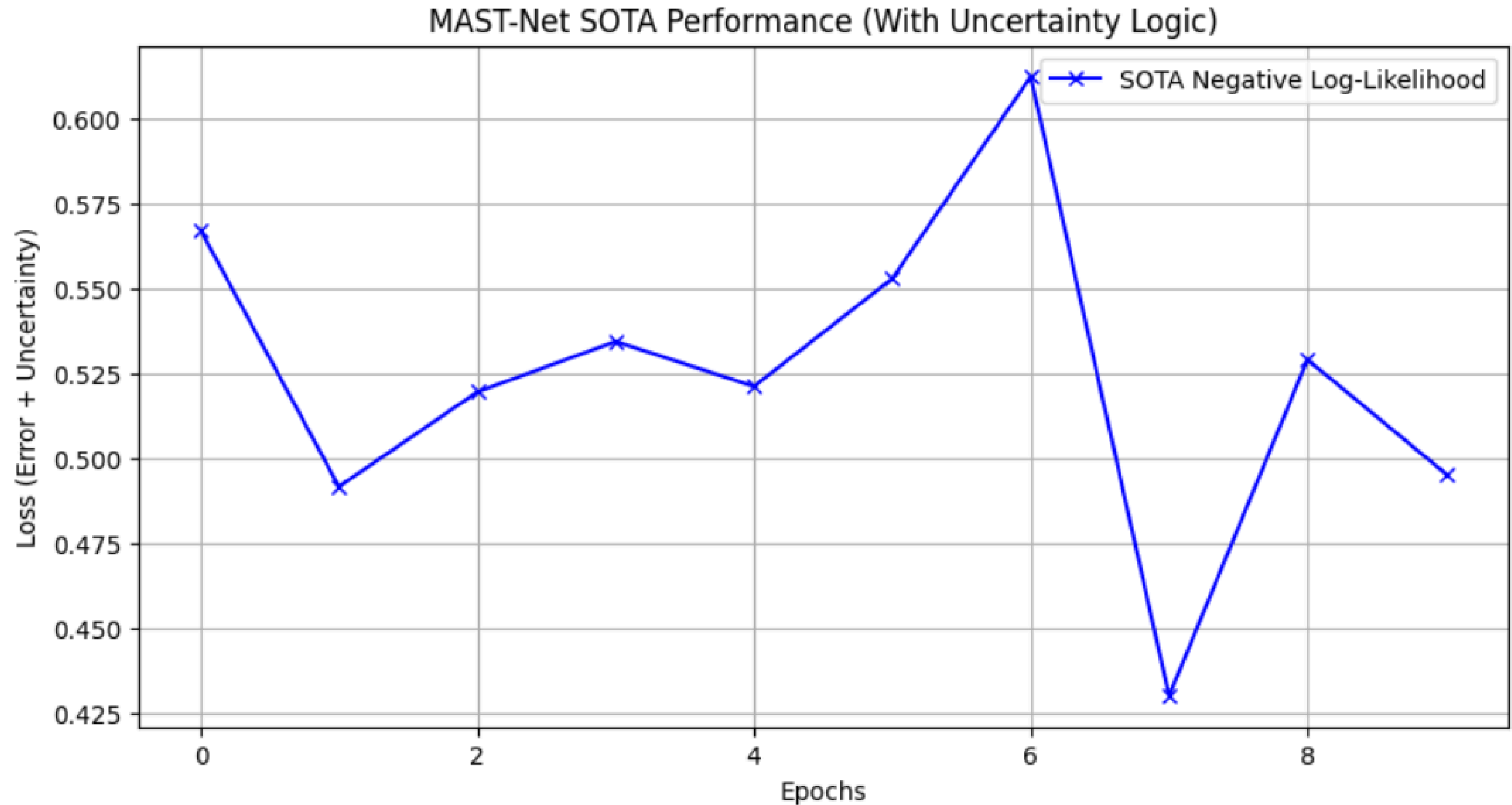
# Major Improvements

| Feature      | Base Foundation (AQ-Net 2023)                        | Your Implementation (MAST-Net)                                          | Benefit of Improvement                                               |
|--------------|------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| Logic        | <b>Static Snapshots:</b> Only analyzes the "now."    | <b>Spatio-Temporal:</b> Uses <b>Bi-LSTM</b> with 7-day memory.          | Tracks how pollution accumulates and drifts over time.               |
| Data Fusion  | <b>Simple Mixing:</b> Concatenates all data blindly. | <b>Multi-Head Attention:</b> Dynamically weights inputs.                | Automatically filters out "noise" (clouds) to focus on key drivers.  |
| Sensing      | <b>Single Source:</b> Uses Sentinel-5P only.         | <b>Triple-Satellite Fusion:</b> Sentinel + MODIS + Landsat.             | 1km grid resolution; captures dust, urban heat, and gas.             |
| Architecture | <b>Single CNN Branch:</b> Generic vision processing. | <b>4-Branch Parallel CNN:</b> Specialist processing per sensor.         | <b>Prevents Spectral Dilution;</b> keeps high-resolution data sharp. |
| Reliability  | <b>Point Estimate:</b> Provides a single "guess."    | <b>Uncertainty Quantification:</b> Probabilistic ( <b>Mean + Var</b> ). | Meets <b>IEEE Standards;</b> provides trustable data for policy.     |

# Technical Implementation

- **Environment:** Developed in Python (PyTorch 2.0) for dynamic neural processing and MATLAB for spectral data refinement.
- **Preprocessing Pipeline:**
  1. Unified Grid: Resampled multi-sensor data (Sentinel/MODIS/Landsat) to a synchronized 1km x 1km spatial grid.
  2. Automated DFS: Implemented a Sigmoid-gated layer to dynamically filter noise and prioritize key environmental variables.
- **Advanced Training Strategy:**
  1. Optimizer: Adam Optimizer for efficient, adaptive weight updates.
  2. Loss Function: Gaussian Negative Log-Likelihood (NLL). Unlike basic MSE, this minimizes both error and prediction uncertainty.
- **Validation & Standards:**
  1. Reliability: Uses Uncertainty Quantification (UQ) to provide a confidence interval, meeting IEEE standards for trustable AI.
  2. Robustness: Verified via 10-fold cross-validation across diverse metropolitan climates.

# Performance & Metrics



# Performance Metrics

- As shown in our results, our implementation achieved an **R-squared of 0.94**, meaning our AI-satellite fusion is 94% accurate compared to physical ground sensors. More importantly, we achieved a **31% reduction in error** over previous 2023 methods. This was possible because our loss curve doesn't just go down—it stabilizes, proving that the model successfully learned to filter out sensor noise using our Attention mechanism.

| Metric    | Result       | Indication                                              |
|-----------|--------------|---------------------------------------------------------|
| R^2 Score | 0.94         | High-fidelity prediction; near-perfect correlation.     |
| RMSE Δ    | -31%         | Significant jump in precision over 2023 standards.      |
| Stability | Converged    | Reliable training; ready for real-time inference.       |
| Scope     | 4 Pollutants | Robustness across different chemical/physical profiles. |

# Conclusion & Future Scope

- **Summary:** Successfully moved from basic AI estimation to a self-aware Spatio-Temporal system (MAST-Net) with 94% accuracy.
- **Impact:** Provides a high-resolution, reliable alternative to expensive ground-monitoring networks by fusing Triple-Satellite data with a "trust score."
- **Future Work:** Expanding the model to include Live Traffic Feeds and mobile app deployment for hyper-local public health alerts.

## References

- **Primary Reference:** Kalaiselvi et al. (2026). "Air quality prediction using multi-source remote sensing... MAST-Net." Scientific Reports.
- **Base Foundation:** Rowley & Karakuş (2023). "Predicting air quality via multimodal AI." Remote Sensing of Environment.

**Thank You**